Principal Components Analysis (PCA):

a) Means and Mean-Centered Data:

$$\left[\bar{X} = \frac{\Sigma x}{n} \right]$$

$$\bar{x} = \frac{4+8+18+7}{4} = 8 \rightarrow (1)$$

$$\bar{y} = \frac{11+4+5+14}{4} = 8.5 \rightarrow (2)$$

Point:	$x - \bar{x}$	$y - \bar{y}$
P_1 =	-4	2.5
P_2	6	-4.5
P_3	5	-3.5
P_4	-1	5.5

b) Covariance Matrix [n-1]:

$$n-1 = 4-1 = 3 \Rightarrow \boxed{n=3}$$

$$\left[\text{Var}(x) = \frac{\Sigma(x-\bar{x})}{n-1}\right]$$

$$\Sigma(x-\bar{x}) = (11+4) + (0) + (13+12) + (0+1) = 42$$

$$\text{Var}(x) = \frac{42}{3} = 14$$

$$\text{Var}(Y) = \frac{(6.25 + 20.25 + 12.25 + 80.25)}{3}$$

$$\text{Var}(Y) = \frac{69}{3} = 23$$

$$\left[\text{Cov}(x,Y) = \frac{\Sigma(x-\bar{x})(Y-\bar{Y})}{n-1}\right]$$

$$= \frac{(-10 - 17.5 - 5.5)}{3} = \frac{-33}{3}$$

$$\left[\text{Cov}(x,Y) = -11\right]$$

c) Eigenvalues and Unit Eigenvectors:

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}, \quad Av = \lambda v$$

$$\left[\det(A - \lambda I) = 0\right]$$

$S \Rightarrow A \Rightarrow \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$

$\lambda = \begin{matrix} a & b \\ +11 & 16.385 \end{matrix}$

$A - \lambda I \Rightarrow$

$$\det \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$= (14-\lambda)(23-\lambda) - 121$

$= \lambda^2 - 37\lambda + 201 = 0$

$\lambda = (37 \pm \sqrt{565})$

$\lambda_1 = 30.385 \quad | \quad \lambda_2 = 6.615$

PCA:

For $\lambda_1 \rightarrow \quad [\Sigma - \lambda I] V = 0$

$(14 - 30.385)a - 11b = 0 \quad \rightarrow (1)$

$-11b = +14a - 30.385a$

$-11b = 16.385\,a$

$b = -\frac{16.385}{11}\,a$

$\boxed{b = -1.4895\,a} \quad \rightarrow (2)$

Let $a = 1$,

Then Vector = 1.7941

$$e = (0.5574, -0.8303)$$

Similar to l_2,

$$e = (0.8303, 0.5574)$$

d) First Principal Component :

$$PC = e_a \cdot x - e_b \cdot Y$$

$$PC = 0.5574\,X - 0.8308\,Y$$

e) 2D into 1D :

Point	Substitution (x, y)	PC Score
P_1	$0.5574(-4) - 0.8303(2.5)$	-4.805
P_2	$0.5574(0) - 0.8303(-4.5)$	3.736
P_3	$0.5574(5) - 0.8303(-3.5)$	5.693
P_4	$0.5574(-1) - 0.8303(9.5)$	-5.124

f) Variance :

$$Var(X) = \frac{d_1}{(d_1 + d_2)} \times 10.0 = \frac{30.385}{30.385 + 6.615}$$

$$= 82.12$$